Infinitely many integer solutions of $2x^2 + xy + 2y^2 + z^3 \pm 1 = 0$

Abstract

We prove, unconditionally and self-containedly, that each of the two Diophantine equations

$$2x^2 + xy + 2y^2 + z^3 - 1 = 0 \qquad \text{and} \qquad 2x^2 + xy + 2y^2 + z^3 + 1 = 0$$

has infinitely many solutions in integers x, y, z. For each sign we exhibit an explicit one-parameter family of polynomial solutions; the verification reduces to two small quadratic identities and Brahmagupta's composition formula for the form $X^2 + 15Y^2$. We then explain how the families were found: the form $2x^2+xy+2y^2$ is the norm form attached to the non-principal ideal class of $\mathbb{Q}(\sqrt{-15})$, and the factorisation $t^3 \pm 1 = (t \pm 1)(t^2 \mp t + 1)$ leads, via a Galois-theoretic analysis, to a conic whose rational points parametrise all candidate quadratic families.

1 Statement

Throughout, $Q(x, y) = 2x^2+xy+2y^2$. It is a positive definite binary quadratic form of discriminant $1 - 16 = -15$, symmetric in x, y, and

$$8\,Q(x, y) = (4x + y)^2 + 15y^2. \tag{1}$$

Since $Q \geq 0$, any solution of $Q(x, y) + z^3 \pm 1 = 0$ has $z \leq 1$, and we write $t = -z$.

Theorem 1. *(A) For every integer s the triple*

$$x = -15s(2s + 1)(4s + 1), \qquad y = 30s^3 + 45s^2 + 15s + 1, \qquad z = -(30s^2 + 15s + 1)$$

satisfies $2x^2 + xy + 2y^2 + z^3 - 1 = 0$.

(B) For every integer s the triple

$$x = -4560s^3-1230s^2+89s+29, \qquad y = 9690s^3+6105s^2+1189s+71, \qquad z = -(570s^2+225s+24)$$

satisfies $2x^2 + xy + 2y^2 + z^3 + 1 = 0$.

In both families z is a strictly decreasing function of $s \geq 0$, so each equation has infinitely many pairwise distinct integer solutions.

The first few members are listed in Table 1.

2 Proof of Theorem 1

The proof uses only (1) and the classical composition identity

$$(A^2 + 15B^2)(C^2 + 15D^2) = (AC - 15BD)^2 + 15(AD + BC)^2, \tag{2}$$

which holds for all A, B, C, D (expand both sides). The idea is to write $t^3 \pm 1 = (t \pm 1)(t^2 \mp t + 1)$, represent each factor (times a power of 2) by $X^2 + 15Y^2$, compose with (2), and read off x, y from (1).

	(A): $Q+z^3-1=0$			(B): $Q+z^3+1=0$		
s	x	y	z	x	y	z
0	0	1	-1	29	71	-24
1	-225	91	-46	-5672	17055	-819
-1	45	1	-16	3270	-4703	-369
2	-1350	451	-151	-41193	104389	-2754
-2	630	-89	-91	31411	-55407	-1854
3	-4095	1261	-316	-133894	320213	-5829

Table 1: Small members of the two families.

Family (A)

Put $t = 30s^2 + 15s + 1$, so $z = -t$ and the claim is $Q(x, y) = t^3 + 1$. One checks directly

$$8(t+1) = 240s^2 + 120s + 16 = 1^2 + 15\,(4s+1)^2, \tag{3}$$

$$4(t^2 - t + 1) = 3600s^4 + 3600s^3 + 1020s^2 + 60s + 4 = (15s^2 + 15s + 2)^2 + 15\,(15s^2 + 7s)^2. \tag{4}$$

(For (4): $(15s^2+15s+2)^2 = 225s^4+450s^3+285s^2+60s+4$ and $15(15s^2+7s)^2 = 3375s^4+3150s^3+735s^2$.) Apply (2) with $A = 1$, $B = 4s+1$, $C = 15s^2+15s+2$, $D = 15s^2+7s$:

$$AC - 15BD = -2\,(450s^3 + 315s^2 + 45s - 1), \qquad AD + BC = 2\,(30s^3 + 45s^2 + 15s + 1),$$

so, dividing $32(t^3+1) = (AC-15BD)^2 + 15(AD+BC)^2$ by 4,

$$8(t^3+1) = X^2 + 15Y^2, \qquad X = -(450s^3 + 315s^2 + 45s - 1), \quad Y = 30s^3 + 45s^2 + 15s + 1. \tag{5}$$

Now $X - Y = -(480s^3 + 360s^2 + 60s) = -60s(8s^2 + 6s + 1) = -60s(2s+1)(4s+1)$ is divisible by 4. Setting $y = Y$ and $x = (X-Y)/4 = -15s(2s+1)(4s+1)$ we get $4x + y = X$, hence by (1) and (5)

$$8\,Q(x, y) = X^2 + 15Y^2 = 8(t^3+1),$$

i.e. $Q(x, y) = t^3 + 1 = -z^3 + 1$, which is $2x^2 + xy + 2y^2 + z^3 - 1 = 0$. □

Family (B)

Put $t = 570s^2 + 225s + 24$, so $z = -t$ and the claim is $Q(x, y) = t^3 - 1$. One checks directly

$$8(t-1) = 4560s^2 + 1800s + 184 = (60s+13)^2 + 15\,(8s+1)^2, \tag{6}$$

$$\begin{aligned} 4(t^2 + t + 1) &= 1299600s^4 + 1026000s^3 + 314220s^2 + 44100s + 2404 \\ &= (285s^2 + 255s + 38)^2 + 15\,(285s^2 + 103s + 8)^2. \end{aligned} \tag{7}$$

(For (6): $(60s+13)^2 = 3600s^2 + 1560s + 169$, $15(8s+1)^2 = 960s^2 + 240s + 15$. For (7): $(285s^2 + 255s+38)^2 = 81225s^4+145350s^3+86685s^2+19380s+1444$ and $15(285s^2+103s+8)^2 = 1218375s^4 + 880650s^3+227535s^2+24720s+960$.) Apply (2) with $A = 60s+13$, $B = 8s+1$, $C = 285s^2+255s+38$, $D = 285s^2 + 103s + 8$:

$$AC-15BD = -2\,(8550s^3-1185s^2-1545s-187), \qquad AD+BC = 2\,(9690s^3+6105s^2+1189s+71),$$

so

$$8(t^3-1) = X^2+15Y^2, \qquad X = -8550s^3+1185s^2+1545s+187, \quad Y = 9690s^3+6105s^2+1189s+71. \tag{8}$$

Here $X-Y = -18240s^3-4920s^2+356s+116 = 4\,(-4560s^3-1230s^2+89s+29)$. With $y = Y$ and $x = (X-Y)/4 = -4560s^3-1230s^2+89s+29$ we have $4x+y = X$, hence $8Q(x,y) = X^2+15Y^2 = 8(t^3-1)$, i.e. $Q(x,y) = t^3-1 = -z^3-1$, which is $2x^2+xy+2y^2+z^3+1 = 0$. □

Both identities (5) and (8), and the final identities $Q(x,y)+z^3 \mp 1 = 0$, were also verified by expanding both sides symbolically; the residuals are identically zero. This completes the proof of Theorem 1.

3 Where the families come from

Nothing in Section 2 depends on what follows; this section records the reasoning that produced the identities, because that is the non-obvious part.

3.1 The form Q as a norm form

Let $K = \mathbb{Q}(\sqrt{-15})$, $\omega = \frac{1+\sqrt{-15}}{2}$, $\mathcal{O}_K = \mathbb{Z}[\omega]$, $\omega^2 = \omega - 4$. The class number of K is 2; the two reduced forms of discriminant -15 are the principal form $P(u,v) = u^2+uv+4v^2 = \mathrm{N}(u+v\omega)$ and Q. Indeed

$$Q(x,y) = \tfrac{1}{2}\,\mathrm{N}(2x+y\omega),$$

and $\{2x+y\omega\} = 2\mathbb{Z}+\omega\mathbb{Z} = \mathfrak{p}_2 = (2,\omega)$ is the non-principal prime above 2. Hence Q represents n exactly when there is an ideal of norm n in the non-trivial class; unwinding this, with χ the class of a prime ($\mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_5$ and the split primes $p \equiv 2, 8 \pmod{15}$ are non-principal, split primes $p \equiv 1, 4 \pmod{15}$ are principal, inert primes $p \equiv 7, 11, 13, 14 \pmod{15}$ must occur to even exponent),

$$Q \text{ represents } n > 0 \iff v_p(n) \text{ even for all inert } p \text{ and } v_2(n)+v_3(n)+v_5(n)+\sum_{p\equiv 2,8\,(15)} v_p(n) \text{ is odd.} \tag{9}$$

This criterion makes it cheap to list all $t \le 2\cdot 10^4$ with $t^3 \pm 1$ represented, which was used to test candidate families. It also gives necessary congruences: values of Q prime to 15 are $\equiv 2, 8 \pmod{15}$, so for $Q = t^3+1$ one needs $t \not\equiv 0 \pmod 3$ and $t \not\equiv 0, 2 \pmod 5$, while for $Q = t^3-1$ one needs $t \not\equiv 2 \pmod 3$ and $t \not\equiv 0, 3 \pmod 5$. (Our families have $t \equiv 1 \pmod{15}$ and $t \equiv 9 \pmod{15}$ respectively.)

In these terms, $8n = X^2+15Y^2$ with $X \equiv Y \pmod 2$ is the same as $2n = \mathrm{N}(\alpha)$ with $\alpha = \frac{X-Y}{2} + Y\omega \in \mathcal{O}_K$, and a short parity argument (replace α by $\bar{\alpha}$ if necessary so that $\alpha \in \mathfrak{p}_2$) shows that this already forces Q to represent n. The proof in Section 2 avoids this lemma by exhibiting x, y directly.

3.2 Why a quadratic $t(s)$, and the Galois constraint

If x, y, z are polynomials in a parameter s with $Q(x,y) = t^3+c$, $c = \pm 1$, then $\deg t = 2d$ and $\deg x = \deg y = 3d$; the simplest case is $d = 1$, $t = as^2+bs+c_0$, and we need a cubic $\xi \in K[s]$ with $\xi\bar{\xi} = 8(t^3+c)$. Over $\overline{\mathbb{Q}}$,

$$t(s)^3 + c = \prod_{j=0}^{2} \big(t(s) + c\rho^j\big), \qquad \rho = e^{2\pi i/3},$$

has six roots $r_1, \dots, r_6$, and ξ corresponds to a triple T of them, stable under $\mathrm{Gal}(\overline{\mathbb{Q}}/K)$, whose complement is its image under any automorphism acting non-trivially on K. Chasing the orbits one finds:

- the two roots of $t(s) = -c$ must lie in $K \setminus \mathbb{Q}$, i.e. $\mathrm{disc}(t+c) = b^2 - 4a(c_0+c) \in -15\,\mathbb{Q}^{\times 2}$;
- the two roots of $t(s) = -c\rho$ must lie in $L = K(\rho) = \mathbb{Q}(\sqrt{-3}, \sqrt{5})$ but not in $\mathbb{Q}(\rho)$, i.e. $\mathrm{disc}(t + c\rho) = E - 4ac\rho \in 5\,\mathbb{Q}(\rho)^{\times 2}$, where $E = b^2 - 4ac_0$;
- the leading coefficient $8a^3$ must be a norm from $K^\times$, equivalently $2a \in \mathrm{N}(K^\times)$.

Equivalently: $t + c$ is (a constant times) a norm of a *linear* polynomial over K, and $t^2 - ct + 1 = \mathrm{N}_{\mathbb{Q}(\rho)/\mathbb{Q}}(t + c\rho)$ is a norm of a *quadratic* polynomial over K, because $t + c\rho$ is itself a norm from L to $\mathbb{Q}(\rho)$ of a linear polynomial. The identities (3)–(4) and (6)–(7) are exactly these two norm statements. (The two conditions "disc $\in \mathbb{Q}^{\times 2}$" or "disc $= 0$" for $t + c$, which look more natural at first, are excluded: they would make ξ rational up to a scalar λ with $\mathrm{N}(\lambda) = 8 \cdot (\text{square})$, and 8 is not a norm from K because $\mathfrak{p}_2^3$ is non-principal.)

3.3 The conic

Write $E - 4ac\rho = 5(g_0 + g_1\rho)^2$. Comparing coefficients of $1, \rho$ (using $\rho^2 = -1 - \rho$),

$$E = 5(g_0^2 - g_1^2), \qquad -4ac = 5g_1(2g_0 - g_1),$$

and then $\mathrm{disc}(t+c) = E - 4ac = 5\big((g_0+g_1)^2 - 3g_1^2\big)$. Requiring this to be $-15m^2$ gives $(g_0+g_1)^2 - 3g_1^2 = -3m^2$; putting $g_0 + g_1 = 3h$ this is the conic

$$g_1^2 - 3h^2 = m^2,$$

with the rational parametrisation $g_1 = 1 + 3\tau^2$, $h = 2\tau$, $m = 1 - 3\tau^2$ (up to a common scalar λ). Substituting back,

$$ac = \tfrac{15}{4}\lambda^2(3\tau^2+1)(3\tau-1)(\tau-1), \qquad E = -60\lambda^2\tau(3\tau^2-3\tau+1), \qquad \mathrm{disc}(t+c) = -15\lambda^2(3\tau^2-1)^2. \tag{10}$$

So every quadratic family is determined by a rational τ and a scaling λ, subject to: $a > 0$, $2a \in \mathrm{N}(K^\times)$, and the existence of integers b, c_0 with $b^2 - 4ac_0 = E$ (a congruence condition modulo $4a$).

Sign -1 (i.e. $c = +1$). Take $\tau = -1$, $\lambda = \frac{1}{2}$: then $ac = \frac{15}{4} \cdot \frac{1}{4} \cdot 4 \cdot (-4)(-2) = 30$, so $a = 30$, and $E = 105$. Here $2a = 60 = P(7,1)$ is a norm. The congruence $b^2 \equiv 105 \pmod{120}$ has the solution $b = 15$, giving $c_0 = (225 - 105)/120 = 1$ and $t = 30s^2 + 15s + 1$. The two norm identities (3), (4) then follow by factoring $t + 1$ and $t^2 - t + 1$ over K (or by the small searches $240 = X^2 + 15Y^2 \Rightarrow (X, Y) = (0, 4)$ and $16 = 1 + 15$).

Sign $+1$ (i.e. $c = -1$). Now $a > 0$ forces $(3\tau - 1)(\tau - 1) < 0$, i.e. $\frac{1}{3} < \tau < 1$, and the norm condition on $2a$ becomes much more restrictive: for $\tau = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{3}{5}, \frac{2}{5}, \frac{4}{5}$ the squarefree part of a contains an inert prime (7, 7, 43, 13, 37, 7 respectively; for $\tau = \frac{4}{5}$ also 73) to an odd power, so these τ are excluded by (9). The first admissible value is $\tau = \frac{3}{7}$ (and $\tau = \frac{7}{9}$ gives the same family), for which the squarefree reduction of (10) gives

$$a = 570 = 2 \cdot 3 \cdot 5 \cdot 19, \qquad E = -4095, \qquad 2a = 1140 = 4 \cdot 285 = \mathrm{N}\big(2(13 + 4\omega)\big),$$

(the choice $\lambda = \frac{49}{2}$ in (10); $285 = P(13,4)$, and indeed $19 \equiv 4 \pmod{15}$ is a principal split prime, so $2a$ has an even number of non-principal prime factors and is a norm by (9)). The congruence $b^2 \equiv -4095 \pmod{2280}$ forces $15 \mid b$, b odd, $b \equiv \pm 60 \pmod{285}$; the solution $b = 225$ gives $c_0 = (225^2 + 4095)/2280 = 24$ and $t = 570s^2 + 225s + 24$. Identities (6), (7) were then obtained by factoring $t - 1$ and $t^2 + t + 1$ over K and rescaling the factors to integral coefficients.

A search over all $\tau = p/q$ with $|p|, q \le 45$ found no quadratic family for the sign $+1$ with a smaller leading coefficient (in the half-integer normalisation of Remark 2 the minimal leading coefficient is $285/2$), which explains why family (B) is unavoidably larger than family (A).

3.4 Two ideas that do not work, for contrast

Making $t^2 \mp t + 1$ a norm first. Solving $(2t \mp 1)^2 + 3 = (2u + v)^2 + 15v^2$ by factoring $3(5v^2 - 1)$ gives the polynomial families $t = 5k^2$ and $t = 15k^2 - 1$ only; but $5k^2$ fails the mod-5 congruence for both signs, and $15k^2 - 1$ fails mod 3 for the sign $+1$. *Making $t^3 \pm 1$ a fixed multiple of a square.* Equations such as $t^3 \pm 1 = cu^2$ are elliptic curves and have only finitely many integer points, so they cannot produce infinitely many solutions. The quadratic-family approach above circumvents both obstacles by letting the two factors of $t^3 \pm 1$ be norms of polynomials of different degrees and composing them.

4 Remarks

Remark 2. Both families extend to half-integral s. Writing $s = u/2$, family (A) has $t = 1 + 15\binom{u+1}{2}$, i.e. $t = 1 + \frac{15u(u+1)}{2}$ for *all* integers u (values $1, 16, 46, 91, 151, 226, \dots$), and family (B) has $t = \frac{285u^2+225u+48}{2}$ (values $24, 54, 279, 369, 819, 969, \dots$). The corresponding (x, y) for odd $u = 2s + 1$ are, respectively,

$$x = -15(s+1)(2s+1)(4s+3), \quad y = 30s^3 + 45s^2 + 15s - 1, \quad z = -(30s^2 + 45s + 16),$$

and

$$x = 4560s^3 + 8070s^2 + 4561s + 804, \quad y = 7410s^3 + 16605s^2 + 12281s + 3001, \quad z = -(570s^2 + 795s + 279);$$

these were verified symbolically in the same way.

Remark 3. Since Q is symmetric and even, $(x, y) \mapsto (y, x)$ and $(x, y) \mapsto (-x, -y)$ map solutions to solutions. Composing the factor representations with the other sign choice in (2) gives further polynomial solutions, e.g. for (A): $x = s(105s^2 + 75s + 13)$, $y = 30s^3 + 30s^2 + 8s + 1$ with the same z.

Remark 4. The families are far from exhaustive: by (9) the equation with sign -1 also has solutions with $z = -8, -14, -23, -31, \dots$ and the one with sign $+1$ with $z = -7, -52, -54, -69, \dots$, none of which lie in the families above. Whether the solutions of either equation can be described by finitely many polynomial families is a different (and harder) question, not addressed here.

Remark 5. No unproved hypothesis is used anywhere: Theorem 1 rests on the polynomial identities (3)–(5) and (6)–(8), each of which can be checked by hand or by any computer algebra system. Section 3 is motivation only.